Appendix IV. Public Debt Sustainability Analysis

The public DSA suggests that Malaysia's public debt remains manageable at under 70 percent of GDP (with gross financing needs below 15 percent of GDP) under different macro-fiscal shocks.

1. Background. The debt sustainability analysis (DSA) framework for market access countries is used to assess Malaysia's debt sustainability and other risks related to its funding and debt structure. The framework uses a risk-based approach and includes: (i) an assessment of the realism of baseline assumptions and the projected fiscal adjustment; (ii) an analysis of risks associated with the debt profile; (iii) macro-fiscal risks; (iv) a stochastic debt projection considering past macro-fiscal volatility; and (v) a standardized summary of risks in a heat map.

2. Macro-fiscal assumptions. Growth is estimated at 5.8 percent in 2017, converging to 4.9 percent in the medium term. In staff's baseline projections, federal government deficit is reduced in the near term from 3.0 percent of GDP in 2017 to 1.5 percent in 2022. The projected fiscal consolidation is consistent with the authorities' targets and is supported by structural reforms announced in recent budgets. The projected consolidation is lower than what is expected under policy commitments.

3. Data coverage. Consistent with the data on government debt reported by the authorities, the fiscal assumptions in the DSA are based on the federal government budget. This coverage excludes local and state governments and statutory bodies which typically borrow from the federal government or receive explicit government guarantees. The liabilities of these entities are therefore captured in the federal government's gross debt and stock of loan guarantees.[1] Borrowing by state owned enterprises, which are in some cases under federal government guarantees, has increased in recent years and is projected to continue to increase in the medium term.

4. Choice of framework. Malaysia's high level of government debt and gross financing requirement calls for using the higher scrutiny framework. Government's gross debt increased sharply in 2009, reflecting sizable discretionary fiscal stimulus, declining real and nominal growth and a large fall in oil prices. Although growth has recovered since then, the primary deficit has remained high, pushing the debt to GDP ratio to about the authorities' debt ceiling of 55 percent. Gross financing needs (GFN) peaked at 10.2 percent of GDP in 2013 and are expected to fall and remain below 8 percent in the medium term.

5. Realism of baseline assumptions. The median forecast error for real GDP growth during 2008–16 is zero, suggesting that there is no evidence of a systemic projection bias that would undermine the assessment. The median forecast error for GDP deflator is -1.8 percent, suggesting that the staff forecasts have been more optimistic. The median forecast error for primary balance suggests that staff projections have been slightly optimistic (a forecast bias of –0.25 percent of GDP), but the forecast bias has improved in the later years.

[1] Gross debt of consolidated general government is not published by the authorities.

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6. Cross-country experience suggests the projected fiscal adjustment is feasible. The maximum three-year adjustment in the cyclically-adjusted primary balance (CAPB) over the projection period (1.5 percent of GDP) is ambitious but realistic. As highlighted earlier, staff does not rule out the existence of implementation risks and therefore considering a no adjustment scenario, as done in this DSA, is necessary to take that into account. Finally, the maximum level of the primary balance (0.1 percent of GDP) that is assumed in the projections is reasonable when compared to the experience in other market-access countries.

7. The DSA framework suggests Malaysia's government debt-to-GDP ratio remains below 70 percent and its gross financing needs remain below 15 percent of GDP under different macroeconomic and fiscal shocks.

- Under the baseline, the debt-to-GDP ratio is projected to decrease to below 45 percent by 2022, but if the projected consolidation does not take place, captured under the constant primary balance simulation, it remains broadly constant at about 50 percent of GDP. Under most macro-fiscal stress tests, debt-to-GDP ratio remains below 60 percent of GDP. If there is a one standard deviation shock to real GDP growth, the debt-to-GDP ratio initially increases about 2 percent by 2020 and declines thereafter. A similar path results in a combined macro-fiscal shock that includes higher interest rates and a lower primary balance and would keep debt at about 50 percent of GDP.

- A permanent oil price shock implies a growing debt-to-GDP profile to about 70 percent in the medium term. The oil price shock in analysis is very large. It assumes a 1 percent of GDP permanent reduction in the oil income. Such a shock would be commensurate with a 33 percent decline in oil price, with an amplifying effect that reduces oil companies' profits by 66 percent.

- A contingent liability shock, whereby the government would have to absorb all government guaranteed loans, totaling 15 percent of GDP, over two years, would increase risks significantly. This shock also includes a persistent shock to growth and interest rate increases. The debt-to-GDP ratio would rise above 55 percent of GDP. Although this is a low probability scenario, the simulations underscore the growing vulnerability posed by contingent liabilities.

- Gross financing needs under all scenarios remain below 10 percent, except for the oil shock scenario in which they grow to almost 15 percent by the end of the projection horizon.

- Stochastic simulations based on historical volatilities in Malaysia's macroeconomic variables also show that the 90th percentile of the debt-to-GDP simulations is below 60 percent.

8. Heat map. Despite the low share of foreign currency and short-term debt in public debt, Malaysia faces risks arising from its external financing requirement and relatively large share of public debt held by foreigners. At 39 percent, the external financing requirement is above the upper threshold of early warning benchmarks and the share of debt held by foreigners is relatively high at about 30 percent of total. As discussed earlier, the existence of large domestic institutional investors who tend to make opportunistic investments is a mitigating factor.

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Figure 1. Malaysia: Public Sector Debt Sustainability Analysis (DSA)—Baseline Scenario

(In percent of GDP unless otherwise indicated)

Debt, Economic and Market Indicators [1/]

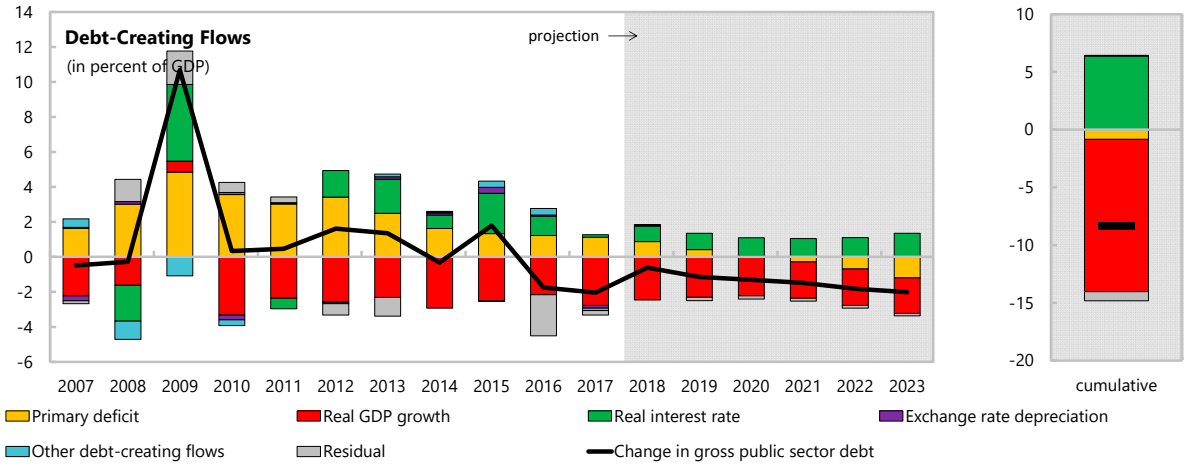
	Actual			Projections					
	2007-2015 [2/]	2016	2017	2018	2019	2020	2021	2022	2023
Nominal gross public debt	48.7	52.7	50.7	50.1	48.9	47.6	46.1	44.3	42.3
Public gross financing needs	9.5	9.9	10.7	7.7	7.0	7.0	7.1	7.1	7.0
Real GDP growth (in percent)	4.8	4.2	5.8	5.3	5.0	4.9	4.7	4.9	4.9
Inflation (GDP deflator, in percent)	2.4	2.0	3.9	3.1	2.8	2.5	2.7	2.6	2.4
Nominal GDP growth (in percent)	7.4	6.3	9.9	8.5	8.0	7.6	7.6	7.6	7.4
Effective interest rate (in percent) [4/]	4.5	4.2	4.5	5.1	5.0	5.0	5.2	5.3	5.8

As of January 02, 2018

Sovereign Spreads		
EMBIG (bp) 3/		122
5Y CDS (bp)		59
Ratings	Foreign	Local
Moody's	A1	A3
S&Ps	A-	A
Fitch	A-	A-

Contribution to Changes in Public Debt

	Actual			Projections							
	2007-2015	2016	2017	2018	2019	2020	2021	2022	2023	cumulative	debt-stabilizing primary balance [9/]
Change in gross public sector debt	1.7	-1.7	-2.0	-0.6	-1.1	-1.3	-1.5	-1.8	-2.0	-8.4	
Identified debt-creating flows	1.4	0.6	-1.8	-0.7	-0.9	-1.1	-1.3	-1.7	-1.9	-7.6	
Primary deficit	2.8	1.2	1.1	0.9	0.4	0.0	-0.3	-0.7	-1.2	-0.8	-0.7
Primary (noninterest) revenue and grants	19.8	17.0	16.5	16.3	16.4	16.5	16.5	16.5	16.6	98.8	
Primary (noninterest) expenditure	22.6	18.2	17.6	17.2	16.8	16.5	16.2	15.8	15.4	97.9	
Automatic debt dynamics [5/]	-1.2	-1.0	-2.8	-1.6	-1.4	-1.2	-1.0	-1.0	-0.7	-6.8	
Interest rate/growth differential [6/]	-1.2	-1.1	-2.6	-1.6	-1.4	-1.2	-1.0	-1.0	-0.7	-6.8	
Of which: real interest rate	0.9	1.1	0.2	0.9	0.9	1.1	1.0	1.1	1.4	6.4	
Of which: real GDP growth	-2.1	-2.2	-2.8	-2.5	-2.3	-2.2	-2.1	-2.1	-2.0	-13.2	
Exchange rate depreciation [7/]	0.0	0.1	-0.2	...	...	...	...	...	...	...	
Other identified debt-creating flows	-0.2	0.4	-0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Please specify (1) (e.g., drawdown of dep	-0.5	-0.1	-0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Contingent liabilities	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Please specify (2) (e.g., ESM and Euroare	0.4	0.5	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Residual, including asset changes [8/]	0.2	-2.4	-0.2	0.0	-0.2	-0.2	-0.2	-0.2	-0.1	-0.8	



Source: IMF staff.

1/ Public sector is defined as the Federal Government.

2/ Based on available data.

3/ Long-term bond spread over U.S. bonds.

4/ Defined as interest payments divided by debt stock (excluding guarantees) at the end of previous year.

5/ Derived as [(r - π(1+g) - g + ae(1+r)]/(1+g+π+gπ)) times previous period debt ratio, with r = interest rate; π = growth rate of GDP deflator; g = real GDP growth rate; a = share of foreign-currency denominated debt; and e = nominal exchange rate depreciation (measured by increase in local currency value of U.S. dollar).

6/ The real interest rate contribution is derived from the numerator in footnote 5 as r - π (1+g) and the real growth contribution as -g.

7/ The exchange rate contribution is derived from the numerator in footnote 5 as ae(1+r).

8/ Includes asset changes and interest revenues (if any). For projections, includes exchange rate changes during the projection period.

9/ Assumes that key variables (real GDP growth, real interest rate, and other identified debt-creating flows) remain at the level of the last projection year.

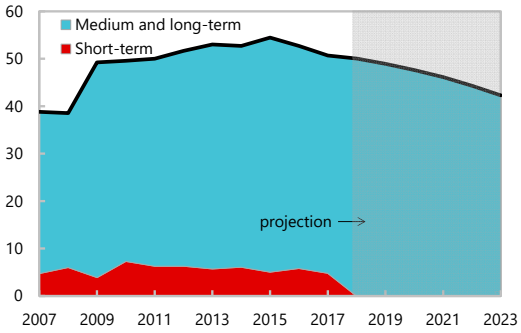
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Figure 2. Malaysia: Public DSA—Composition of Public Debt and Alternative Scenarios

Composition of Public Debt

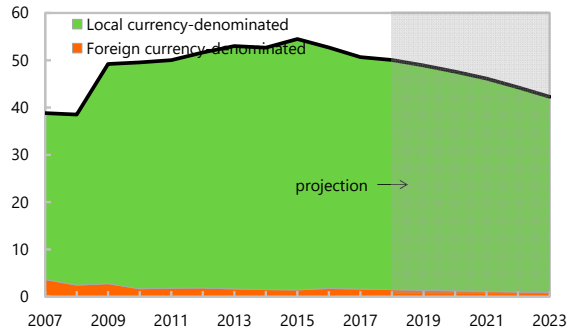
By Maturity

(in percent of GDP)

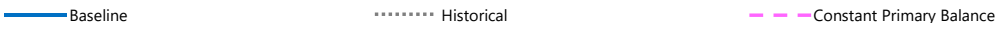


By Currency

(in percent of GDP)

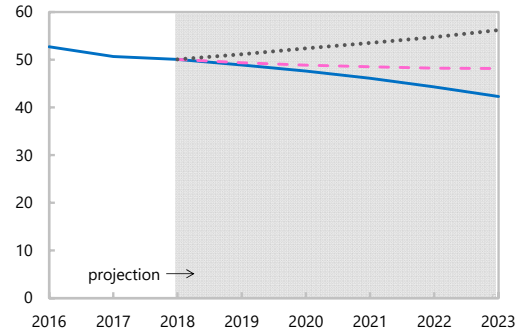


Alternative Scenarios



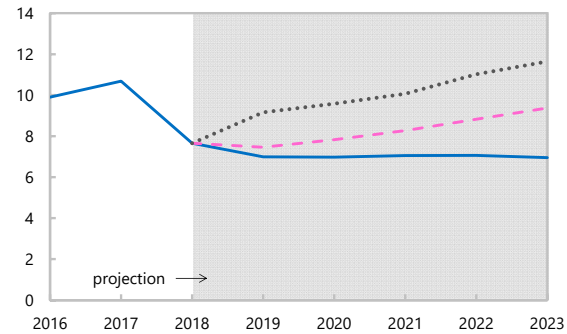
Gross Nominal Public Debt

(in percent of GDP)



Public Gross Financing Needs

(in percent of GDP)



Underlying Assumptions

(in percent)

Baseline Scenario	2018	2019	2020	2021	2022	2023
Real GDP growth	5.3	5.0	4.9	4.7	4.9	4.9
Inflation	3.1	2.8	2.5	2.7	2.6	2.4
Primary Balance	-0.9	-0.4	0.0	0.3	0.7	1.2
Effective interest rate	5.1	5.0	5.0	5.2	5.3	5.8
Constant Primary Balance Scenario						
Real GDP growth	5.3	5.0	4.9	4.7	4.9	4.9
Inflation	3.1	2.8	2.5	2.7	2.6	2.4
Primary Balance	-0.9	-0.9	-0.9	-0.9	-0.9	-0.9
Effective interest rate	5.1	5.0	5.0	5.2	5.3	5.7

Historical Scenario	2018	2019	2020	2021	2022	2023
Real GDP growth	5.3	4.7	4.7	4.7	4.7	4.7
Inflation	3.1	2.8	2.5	2.7	2.6	2.4
Primary Balance	-0.9	-2.6	-2.6	-2.6	-2.6	-2.6
Effective interest rate	5.1	5.0	4.9	5.0	5.1	5.4

Source: IMF staff.

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